PROBLEM 13 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

Characterisation of an Orthorhombic Metal: (220) Plane and [101] Direction

Original statement · Problems T2_CrystStruct.pdf, p. 3

13. In the figure, the dimensions of the (220) ***plane and the*** $[101]$ ***direction of a hypothetical metal with a radius of*** 1.98 Å ***and atomic weight of*** 180.95 g/mol ***are shown.*** *Plane* (220)*: Width* $=$ 3.9 Å*, Height* $=$ 7.0 Å*. Direction* $[101] = 8.6$ Å *(exhibiting 3 atoms in contact along the diagonal, so length* $= 4R$*).* ***a)*** *Draw the unit cell, deduce which crystal system it belongs to, and identify the name of this crystal structure.* ***b)*** *Determine the packing factor.* ***c)*** *Calculate the density on the* (110) *and* (100) *planes in* atoms/cm^2*.* ***d)*** *Calculate the interplanar distances* $d_{(111)}$ *and* $d_{(110)}$ *in* Å*.* ***e)*** *Calculate the weight of a single crystal with a volume of* $1\ \text{cm}^3$*. (Solution: a) Face centred Orthorhombic; b)* 0.62*; c)* $\rho_{(110)} = 3.66 \times 10^{14}\ \text{atoms/cm}^2$*,* $\rho_{(100)} = 4.76 \times 10^{14}\ \text{atoms/cm}^2$*; d)* $d_{(111)} = 3.37$ Å*,* $d_{(110)} = 3.84$ Å*; e)* 5.72 g*)*

Solution: a) Face centred Orthorhombic; b) 0.62**; c)** $\rho_{(110)} = 3.66 \times 10^{14}\ \text{atoms/cm}^2$**,** $\rho_{(100)} = 4.76 \times 10^{14}\ \text{atoms/cm}^2$**; d)** $d_{(111)} = 3.37$ Å**,** $d_{(110)} = 3.84$ Å**; e)** 5.72 g

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Atomic radius: $R = 1.98\ \text{Å} = 1.98 \times 10^{-8}$ cm.
- Molar atomic weight: $M = 180.95$ g/mol (Tantalum, Ta).
- Height of the (220) plane: $c = 7.0$ Å.
- Width of the (220) plane: in an orthogonal crystal, the (220) plane cuts at $a/2$ and $b/2$, so its width is half the base diagonal:

$$\frac{\sqrt{a^2 + b^2}}{2} = 3.9\ \text{Å} \implies \sqrt{a^2 + b^2} = 7.8\ \text{Å}$$

- Length of the $[101]$ direction:

$$L_{[101]} = \sqrt{a^2 + c^2} = 8.6 \text{ Å} \approx 4R = 4(1.98) = 7.92 \text{ Å}$$

- Avogadro constant: $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Deduction of the Lattice Parameters (a, b, c):**
2. We know $c = 7.0$ Å.
3. From the diagonal $[101]$: $a^2 + c^2 = 8.6^2 \implies a^2 = 8.6^2 - 7.0^2 = 73.96 - 49.00 = 24.96 \text{ Å}^2 \implies a \approx 5.0$ Å.
4. From the base diagonal: $a^2 + b^2 = 7.8^2 = 60.84 \implies b^2 = 60.84 - 24.96 = 35.88 \text{ Å}^2 \implies b \approx 6.0$ Å.
5. Since $a \neq b \neq c$ ($a \approx 5.0$ Å, $b \approx 6.0$ Å, $c = 7.0$ Å) and the angles are 90°, the system is **Orthorhombic** [Session 3 Slide 13].
6. The atomic contacts along the face diagonals reveal a **Face-Centred (F)** lattice with $n = 4$ equivalent atoms.
7. **Atomic Packing Factor (APF):**

$$\text{APF} = \frac{n \cdot \left(\frac{4}{3}\pi R^3\right)}{a \cdot b \cdot c}$$

8. **Planar Densities:**

$$\rho_{(110)} = \frac{2}{\sqrt{a^2 + b^2} \cdot c}, \quad \rho_{(100)} = \frac{2}{b \cdot c}$$

9. **Interplanar Spacings in an Orthorhombic Lattice [Session 3 Slide 53]:**

$$d_{hkl} = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}}$$

10. **Mass of a Single Crystal of $V = 1 \text{ cm}^3$:**

$$\rho = \frac{n \cdot M}{V_C \cdot N_A}, \quad m = \rho \cdot V$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Part a: Parameters and Lattice Type

- $c = 7.0$ Å.
- $a = \sqrt{8.6^2 - 7.0^2} = \sqrt{73.96 - 49.00} = \sqrt{24.96} = 4.996$ Å $\approx \mathbf{5.0}$ Å.
- $b = \sqrt{7.8^2 - 24.96} = \sqrt{60.84 - 24.96} = \sqrt{35.88} = 5.990$ Å $\approx \mathbf{6.0}$ Å.
- Since $a \neq b \neq c$ and the axes are orthogonal, the crystal system is **Orthorhombic**.
- Having atoms at all face centres, it is the **Face-Centred Orthorhombic Lattice (Face-Centered Orthorhombic)** with $n = 4$.

---

2. Part b: Packing Factor (APF)

- Unit cell volume:

$$V_C = a \cdot b \cdot c = 4.996 \times 5.990 \times 7.0 = 209.48 \text{ Å}^3$$

- Volume of the 4 atoms ($R = 1.98$ Å):

$$V_{\text{atoms}} = 4 \times \left(\frac{4}{3}\pi R^3\right) = \frac{16}{3}\pi(1.98)^3 = \frac{16}{3}\pi(7.7624) = 130.06 \text{ Å}^3$$

- Packing factor:

$$\text{APF} = \frac{130.06}{209.48} = \mathbf{0.6209} \approx \mathbf{0.62}$$

---

3. Part c: Planar Densities

- (110) **Plane:**
- Dimensions: base $\sqrt{a^2 + b^2} = 7.8$ Å $= 7.8 \times 10^{-8}$ cm; height $c = 7.0$ Å $= 7.0 \times 10^{-8}$ cm.
- Area: $A_{(110)} = 7.8 \times 7.0 = 54.6 \text{ Å}^2 = 5.46 \times 10^{-15}$ cm^2.
- Atoms contained: $N_{\text{atoms}} = 2$.

$$\rho_{(110)} = \frac{2 \text{ at}}{5.46 \times 10^{-15} \text{ cm}^2} = 3.663 \times 10^{14} \approx \mathbf{3.66 \times 10^{14}} \text{ atoms/cm}^\mathbf{2}$$

- (100) **Plane:**

- Dimensions: base $b = 5.990\ \text{Å} = 5.990 \times 10^{-8}\ \text{cm}$; height $c = 7.0\ \text{Å} = 7.0 \times 10^{-8}\ \text{cm}$.
- Area: $A_{(100)} = 5.990 \times 7.0 = 41.93\ \text{Å}^2 = 4.193 \times 10^{-15}\ \text{cm}^2$.
- Atoms contained: $N_{\text{atoms}} = 2$.

$$\rho_{(100)} = \frac{2\ \text{at}}{4.193 \times 10^{-15}\ \text{cm}^2} = 4.770 \times 10^{14} \approx \mathbf{4.77 \times 10^{14}}\ \text{atoms/cm}^2$$

---

4. Part d: Interplanar Distances

- **Spacing $d_{(111)}$:**

$$\frac{1}{d_{(111)}^2} = \frac{1^2}{a^2} + \frac{1^2}{b^2} + \frac{1^2}{c^2} = \frac{1}{24.96} + \frac{1}{35.88} + \frac{1}{49.00}$$

$$\frac{1}{d_{(111)}^2} = 0.040064 + 0.027871 + 0.020408 = 0.088343\ \text{Å}^{-2}$$

$$d_{(111)} = \frac{1}{\sqrt{0.088343}} = \frac{1}{0.297225} = \mathbf{3.364}\ \text{Å} \approx \mathbf{3.36}\ \text{Å}$$

- **Spacing $d_{(110)}$:**

$$\frac{1}{d_{(110)}^2} = \frac{1^2}{a^2} + \frac{1^2}{b^2} + 0 = \frac{1}{24.96} + \frac{1}{35.88} = 0.040064 + 0.027871 = 0.067935\ \text{Å}^{-2}$$

$$d_{(110)} = \frac{1}{\sqrt{0.067935}} = \frac{1}{0.26064} = \mathbf{3.837}\ \text{Å} \approx \mathbf{3.84}\ \text{Å}$$

---

5. Part e: Mass of the $1\ \text{cm}^3$ Single Crystal

- **Cell volume:** $V_C = 209.48\ \text{Å}^3 = 2.0948 \times 10^{-22}\ \text{cm}^3$.
- **Theoretical density:**

$$\rho = \frac{4 \times 180.95\ \text{g/mol}}{(2.0948 \times 10^{-22}\ \text{cm}^3) \times (6.022 \times 10^{23}\ \text{mol}^{-1})} = \frac{723.8}{126.15} = \mathbf{5.738}\ \text{g/cm}^3 \approx \mathbf{5.74}\ \text{g/cm}^3$$

- For a macroscopic volume $V = 1.0\ \text{cm}^3$:

$$m = \rho \cdot V = 5.74\ \text{g/cm}^3 \times 1.0\ \text{cm}^3 = \mathbf{5.74}\ \text{g}$$

Discrepancy with the official solution. The official key ($\rho_{(100)} = 4.76 \times 10^{14}$, $d_{(111)} = 3.37$ Å, $m = 5.72$ g) is reproduced only if the cell is first rounded to $a = 5.0, b = 6.0, c = 7.0$ Å ($V = 210$ Å^3 : $\rho = 5.723\ \text{g/cm}^3$, $d_{(111)} = 3.367$ Å). Without that rounding, the cell obtained from the figure data ($a = 4.996, b = 5.990, c = 7.0$ Å) gives $\rho_{(100)} = 4.77 \times 10^{14}\ \text{atoms/cm}^2$, $d_{(111)} = 3.36$ Å and $m = 5.74$ g (0.3-0.4% differences, within the two-significant-figure precision of the figure). The unrounded values are reported.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- The results agree with the official solution except for rounding differences (see the discrepancy note): $\text{APF} = 0.62$ reflects the lower packing typical of an anisotropic orthorhombic cell relative to the FCC limit (0.74), and the planar densities confirm that the most densely packed plane (100) has a smaller area and a higher atomic concentration than (110).